

GREENLOOP ROBOTICS / INTERNAL KNOWLEDGE PACK

Sorter X1 Product Specification

Controlled technical requirements for hardware, AI, cloud, security, service, and acceptance

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DOCUMENT CONTROL

Document status and use

This specification defines the controlled baseline for the GreenLoop Sorter X1 Rev B and its cloud services. Contract-specific requirements apply only when explicitly referenced in an approved order.

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This document describes a fictional company and fictional performance data. It is intentionally written as realistic internal material for retrieval, grounding, fact-checking, and citation exercises.

Version history

Version	Date	Summary	Approved by
1.0	10 Jan 2025	Initial controlled release	Executive Committee
2.1	15 June 2025	Current release with FY2025 operating updates	Product Engineering

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Section numbers are stable anchors for retrieval and human review.

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SECTION 1

Product purpose and approved use cases

The GreenLoop Sorter X1 is an indoor, attended waste-sorting station that identifies a single deposited item, predicts one of five material classes, and routes the item to a corresponding internal bin.

Approved environments

- University and corporate campuses.
- Airport and transport waiting areas outside security-critical baggage flows.
- Shopping centers, food courts, and event venues.
- Hospitals in public areas for ordinary non-clinical waste only.
- Municipal demonstration and education sites.

Primary outcomes

The system aims to improve separation consistency, measure material volumes, reduce manual inspection, and provide operators with fill, health, and classification analytics. It is not a hazardous-waste identifier and does not certify regulatory compliance of the customer's waste stream.

Intended operator

Routine use requires no technical training. Installation, guard removal, jam clearing inside the mechanism, calibration, and electrical service are restricted to trained personnel.

The X1 is designed for one ordinary item per deposit. It is not approved for medical, chemical, hot, sharp, pressurized, radioactive, or industrial hazardous waste.

SECTION 2

Configurations and commercial variants

All X1 variants share the Rev B cabinet and GL-Vision 3.4 classification baseline. Differences concern cloud services, identity integration, support, and optional modules.

Variant	Included capability	Typical customer
X1 Standard	Local interface, five-bin routing, basic cloud health	Single-site deployments
X1 Pro	Advanced analytics, API, SSO, 99.5% dashboard SLA	Multi-site organizations
X1 Education	Transparent demo panels and learning mode	Schools and public exhibitions

Optional modules

- Twenty-minute internal UPS.
- Receipt or reward QR module.
- Large-format 19-inch display.
- Local cellular gateway.
- External fullness beacon.
- Customer-branded enclosure panels.

Configuration control

The unit serial number links to hardware revision, firmware, model version, options, destination, warranty, and commissioning record. A customer-facing name does not replace the controlled configuration record.

SECTION 3

System architecture

The X1 separates sensing, classification, routing, safety, and cloud functions so that a cloud interruption does not prevent ordinary local sorting.

Layer	Primary components	Failure behavior
Interaction	Deposit door, indicator ring, display	Prompts user to wait or retry
Sensing	RGB camera, time-of-flight, weight, door sensors	Rejects uncertain cycle
Edge intelligence	GL Edge controller and GL-Vision model	Continues offline with cached policy
Routing	Rotary diverter, position encoder, bin doors	Stops safely on mismatch
Safety	Interlocks, emergency stop, overcurrent protection	Removes motion command
Cloud	Telemetry, fleet health, analytics, user management	Queues local events for later upload

Control principle

A sort cycle is accepted only when item presence, door state, classification confidence, diverter position, and receiving-bin availability meet the active policy. The controller records reason codes for rejected or fallback cycles.

Offline operation

The unit can classify and route for up to 72 hours without cloud connectivity. Dashboard data, remote commands, and fleet configuration changes are unavailable until the connection is restored.

SECTION 4

Physical dimensions and materials

The Rev B cabinet is designed for indoor public use and service access from the rear. Dimensions exclude optional signage and required service clearance.

Characteristic	Specification
Overall size	1180 mm wide x 760 mm deep x 1760 mm high
Empty mass	186 kg without removable bins
Installed mass	Up to 245 kg depending on configuration
Frame	Powder-coated galvanized steel
User surfaces	Flame-retardant ABS and stainless steel contact trim
Service clearance	800 mm rear and 500 mm on each side
Anchoring	Four M10 anchors into verified substrate

Ingress and cleaning

The cabinet is splash resistant for normal indoor cleaning but is not assigned an outdoor IP rating. High-pressure water, steam, hose cleaning, and floor flooding can damage sensors, seals, and electrical assemblies.

Finish

Standard colors are GreenLoop Deep Green and neutral gray. Customer graphics may not cover ventilation, labels, cameras, door edges, emergency controls, or service access.

SECTION 5

Power and environmental limits

Site conditions strongly influence reliability and classification performance. The installer records voltage, protective earth, temperature, lighting, network, and floor conditions during commissioning.

Parameter	Required range	Notes
Input power	110 to 240 VAC, 50/60 Hz	Dedicated grounded outlet
Typical consumption	95 W	Idle with cloud connected
Peak consumption	250 W	During routing and display peak
Operating temperature	5 C to 40 C	Indoor conditioned environment
Relative humidity	20% to 80%	Non-condensing
Ambient light	300 to 1200 lux	Avoid direct glare into camera
Floor slope	Less than 2 degrees	Rigid, level, anchored surface
Acoustic level	Below 60 dBA at 1 m	Normal sort cycle

Power loss

Without the optional UPS, a power loss stops the mechanism and preserves the event journal. The optional UPS supports controlled shutdown and up to 20 minutes of telemetry and non-routing operation.

Classification outside the specified lighting range may remain functional but is not covered by the laboratory accuracy specification.

SECTION 6

Sensors and item detection

The sensing chamber combines visual, distance, weight, door, and position signals. No single sensor is treated as sufficient evidence that a valid item is ready for classification.

Sensor	Purpose	Nominal capability
RGB camera	Material and appearance features	1920 x 1080 at controlled exposure
Time-of-flight	Item envelope and chamber occupancy	20 to 600 mm working range
Load cell	Item presence and gross mass	20 g to 2.5 kg operating range
Ultrasonic fill sensors	Remaining bin capacity	Five independent channels
Door interlocks	Safe state confirmation	Dual-channel service-door monitoring
Diverter encoder	Target position confirmation	Absolute position with tolerance check

Capture sequence

After the user closes the deposit door, the controller verifies stability, captures a short image burst, selects the best frame, obtains weight and envelope features, and then invokes the model. Unstable or oversized items are rejected before routing.

Calibration

Camera exposure and reference color are checked during commissioning and after camera replacement. Load-cell zero is checked daily and calibrated quarterly or after mechanical work.

SECTION 7

Edge compute and software baseline

The edge controller hosts the real-time state machine, classification runtime, local database, health monitor, and secure cloud connector.

Component	Controlled baseline
Edge compute	8 GB industrial controller with 12 TOPS inference accelerator
Storage	128 GB encrypted industrial SSD
Operating system	GreenLoop Edge OS 5.2 LTS
Controller firmware	X1 Control 2.8.3
Model runtime	GL Runtime 3.4
Local event buffer	Minimum 72 hours at rated daily volume
Update method	Signed A/B image with automatic rollback

Update policy

Security fixes may be applied during the customer's approved maintenance window. Model updates are staged to a pilot group, monitored against quality gates, and expanded only after Product Quality approval.

Rollback

A failed health check after update returns the unit to the last known good image. Local maintenance may force rollback only with an authorized service credential and a recorded ticket.

SECTION 8

Classification model and supported classes

GL-Vision 3.4.1 predicts five operational classes. The class describes the route configured for the site; it is not a chemical analysis of material composition.

Class	Representative items	Important exclusions
Paper	Office paper, clean cartons, paper sleeves	Wet or heavily food-soiled fiber
Plastic	Bottles, cups, rigid containers	Film tangles and mixed laminates
Metal	Aluminum cans, small clean tins	Batteries and pressurized containers
Organic	Food scraps and compostable residue	Clinical or animal waste
Other	Unknown, mixed, low-confidence items	Does not imply safe general disposal

Confidence policy

A site policy maps prediction and confidence to a route. Predictions below the approved confidence threshold route to Other or request user retry. Customers may not lower the threshold below the controlled minimum without a documented pilot.

Model scope

The production model was trained and validated on ordinary indoor consumer waste. It is not designed for industrial scrap, medical waste, powders, liquids without containers, or objects larger than the input envelope.

SECTION 9

Laboratory performance specification

Laboratory metrics were measured on the controlled GL-X1-B validation set using single, clean, non-overlapping items, specified lighting, a clean lens, and calibrated sensors.

Metric	Specification	Test condition
Macro classification accuracy	93.2%	Five-class controlled validation set
Paper accuracy	96.1%	Clean representative items
Plastic accuracy	94.0%	Rigid plastic mix
Metal accuracy	95.2%	Cans and small tins
Organic accuracy	91.0%	Representative food waste
Other accuracy	89.7%	Mixed approved unknown set
P95 decision time	1.8 seconds or less	From stable capture to route decision

Interpretation

The 93.2% result is a laboratory macro accuracy, not a guarantee for every item or customer site. Field accuracy depends on waste mix, lighting, cleanliness, user behavior, installation, maintenance, and active software.

Do not compare a field metric with the 93.2% laboratory value without describing the different conditions.

SECTION 10

Input rules and multi-item handling

The X1 accepts one item at a time. The chamber uses visual and depth checks to detect obvious multiple objects, but reliable separation of overlapping or bundled items is outside the controlled product claim.

Accepted envelope

Parameter	Limit
Maximum item dimensions	350 x 250 x 200 mm
Maximum item mass	2.5 kg
Minimum reliable mass	20 g
Recommended spacing	5 seconds between users
Deposit rule	One unbagged item per cycle

Multiple items

When multiple objects are detected with high confidence, the user is prompted to remove and reinsert one item. Overlapping items can appear as a single mixed object. If not detected, they may be routed to Other or occasionally to an incorrect material bin.

Bags and bundles

Closed bags, nested cups, compressed bundles, strings, and long flexible film are not supported. These inputs can hide material, exceed the envelope, wrap around moving parts, or create misleading visual features.

Known limitation: multi-item classification errors are possible. The product specification does not assign a financial loss value to those errors.

SECTION 11

Mechanical routing and safety

A rotary diverter aligns the sensing chamber with one of five receiving bins. Motion is allowed only when covers and service doors are closed, the path is clear, and a valid target position is available.

Cycle sequence

- Lock the user deposit door.
- Confirm stable item and safe interlock state.
- Classify and select receiving bin.
- Move diverter and verify encoder position.
- Release item and confirm chamber clear.
- Return diverter to ready position and unlock deposit door.

Safety functions

Function	Behavior
Emergency stop	Removes motion command and requires manual reset
Service-door interlock	Prevents automatic motion while open
Motor overcurrent	Stops motion and records jam fault
Position mismatch	Stops and blocks new deposits
Restart after power loss	Requires self-test before user-ready state

Jam clearing

Only trained service personnel may open the guarded mechanism. Energy isolation is required before hands or tools enter the routing volume.

SECTION 12

Bin capacity and fill monitoring

Five removable 60-liter bins provide a nominal total volume of 300 liters. Actual usable volume depends on item shape, compaction, liners, and local waste practice.

Fill estimation

Each bin uses an ultrasonic distance channel and a model that converts several readings into an estimated percentage. Readings are filtered to reduce false alerts from irregular item surfaces.

Threshold	Default action
70%	Dashboard advisory
85%	Service notification
95%	High-priority collection request
Full or sensor blocked	Route disabled or redirected by site policy

Service access

Bins are removed from the rear service area. The unit must be placed in service mode before a bin door is opened. Operators verify liner position and sensor clearance after replacement.

Accuracy

Fill percentage is an operational estimate with a typical error of plus or minus 8 percentage points for ordinary mixed shapes. It is not a trade-certified volume or weight measurement.

SECTION 13

User interface and accessibility

The interface guides users through deposit, wait, retry, success, and unavailable states using text, color, icon, and sound so that color is not the only signal.

Standard interaction

- Idle prompt invites one item.
- Door-open state shows placement guidance.
- Processing state asks the next user to wait.
- Result state displays predicted class and destination color.
- Retry state explains multiple, oversized, or uncertain input.
- Unavailable state directs the user to an ordinary waste station.

Accessibility

Default text meets the product contrast target, essential controls are reachable from the front approach, sound can be adjusted or disabled, and icons accompany class colors. Site signage and physical approach remain the customer's responsibility.

Languages

The standard release supports English and Arabic. Additional language packs require translation review, layout verification, and customer approval before deployment.

SECTION 14

GreenLoop Cloud dashboard

The cloud dashboard provides fleet status, fill levels, classification trends, maintenance planning, site comparison, user management, and export functions. It does not control safety functions.

Primary modules

Module	Key functions
Fleet overview	Connectivity, health, fill state, software version
Material analytics	Counts, estimated volume, class mix, trend comparison
Quality review	Authorized review of sampled low-confidence events
Maintenance	Tasks, faults, service history, parts
Administration	Sites, users, roles, SSO, API keys
Reports	Scheduled PDF and CSV summaries

Data freshness

Connected units normally deliver health telemetry within five minutes and event records within fifteen minutes. Offline units queue records and mark the last successful contact time.

Commands

Approved users can request diagnostics, schedule updates, change non-safety site policy, and place a unit in maintenance mode. Safety interlocks and emergency-stop behavior cannot be disabled from the cloud.

SECTION 15

Security architecture

Security uses device identity, encrypted communication, signed software, least privilege, tenant separation, audit logging, and monitored cloud services.

Control	Baseline
Data in transit	TLS 1.3 for supported cloud connections
Data at rest	AES-256 or equivalent managed encryption
Device identity	Unique certificate stored in hardware-backed key storage
User authentication	MFA; SAML or OIDC SSO in Pro
Authorization	Role-based access with tenant and site scope
Software integrity	Signed updates with verified boot chain
Audit retention	Minimum 24 months for administrative events
Vulnerability handling	Risk-ranked remediation and customer advisory process

Secrets

API secrets and device credentials are not displayed after creation. Customer administrators can revoke user sessions and rotate API keys. GreenLoop support does not request passwords through email or chat.

Shared responsibility

Customers control user approval, SSO settings, site network rules, physical access, exports, and local retention choices. GreenLoop controls platform operation, device software signing, tenant separation, and service monitoring.

SECTION 16

Data privacy and retention

The platform minimizes image use while retaining enough evidence to evaluate low-confidence behavior and investigate support cases. Customer agreements and applicable law may impose stricter limits.

Data type	Default retention	Customer configuration
Routine event metadata	24 months	12 to 36 months by plan
Sampled classification image	30 days	7 to 90 days with approval
Low-confidence review image	30 days	7 to 90 days with approval
Device health logs	12 months	Contract dependent
Administrative audit log	24 months	Longer export available
Support case attachment	Case term plus 90 days	Deletion request subject to legal hold

Image capture

The camera is directed into the deposit chamber. Customers must position the unit and signage to reduce capture of bystanders. Images are used only for configured operational purposes and are not used for facial recognition.

Export

Image export is disabled by default and restricted to authorized quality roles. Exports create an audit record and must be stored only in approved systems.

Default sampled image retention is 30 days. Employee policy separately prohibits storing customer images on personal devices.

SECTION 17

Network and deployment requirements

A wired Ethernet connection is preferred. Wi-Fi or cellular may be used when signal, security, data allowance, and local policy meet the site design.

Requirement	Minimum
Bandwidth	2 Mbps down and 1 Mbps up average per unit
Latency	Less than 250 ms preferred for dashboard responsiveness
Addressing	DHCP reservation or documented static configuration
Outbound access	HTTPS 443 and approved time synchronization
Inbound access	No public inbound port required
Time	Authenticated or approved NTP source
Segmentation	Dedicated IoT or managed device network recommended

Proxy and inspection

TLS interception can break certificate-based device authentication and requires engineering review. The customer provides proxy details and certificates before commissioning.

Network loss

Sorting continues locally for up to 72 hours. The unit displays a service indicator after the configured connectivity threshold and uploads queued events when service returns.

SECTION 18

API and external integrations

The Pro API supports approved integrations with facilities, sustainability, help desk, and analytics systems. It is not a real-time safety interface.

Core resources

- Sites and units.
- Health and connectivity state.
- Fill-level observations.
- Aggregated classification events.
- Maintenance tasks and fault codes.
- User-managed webhooks for selected status changes.

Limits

Control	Default
API rate	120 requests per minute per tenant
Webhook retry	Exponential retry for up to 24 hours
Export job	Maximum 12 months per request
API key lifetime	Maximum 180 days unless contract specifies less
Image access	Not available through the general API

Integration responsibility

Customers protect credentials, validate received data, handle duplicate webhooks, and respect rate limits. GreenLoop may disable an integration that threatens service stability or security.

SECTION 19

Roles, permissions, and audit records

Permissions are assigned by role and scope. Customer administrators should grant the minimum site and function required.

Role	Typical permissions
Viewer	Read fleet, fill, and standard analytics
Site operator	Viewer plus service acknowledgement and collection tasks
Quality reviewer	Authorized sampled-image review and labels
Maintenance technician	Diagnostics, service mode, work records
Tenant administrator	Users, sites, SSO, API keys, retention policy
GreenLoop support	Time-bound support access approved and logged

Audit events

The system records sign-in, failed authentication, role change, API key creation, retention change, export, support access, configuration change, and remote command events with actor, time, tenant, target, and outcome.

Access review

Tenant administrators review access at least quarterly and after personnel or supplier changes. Dormant accounts may be disabled automatically after the contracted inactivity period.

SECTION 20

Site readiness and installation

Installation begins only after the customer and GreenLoop confirm structural, electrical, network, access, cleaning, waste, and safety readiness.

Readiness checklist

- Level structural floor and approved anchor points.
- Dedicated grounded outlet within cable reach.
- Network path tested with required destinations.
- Specified lighting without direct glare.
- Rear and side service clearances available.
- Waste collection plan and bin liners ready.
- Public flow does not block emergency egress.
- Named customer administrator and site operator available.

Installation

Trained installers inspect shipping condition, move and anchor the cabinet, connect power and network, verify protective earth, install bins, load the approved configuration, and perform safety and functional tests.

Handover

The customer receives serial records, site photos, acceptance results, operator training, admin access, maintenance schedule, and open-item list. A unit with a failed safety test is not handed over for public use.

SECTION 21

Commissioning and acceptance

Commissioning confirms that the delivered configuration works at the actual site. It does not repeat the complete product validation program.

Acceptance test	Minimum result
Safety interlocks	All service doors stop or prevent motion
Emergency stop	Motion removed and reset required
Routing	Ten successful cycles to each configured bin
Classification sample	At least 45 of 50 approved site items routed correctly
Connectivity	Telemetry and command acknowledgement received
Fill sensors	Each channel responds to reference target
User interface	Selected language, sound, and signage verified

Exceptions

An acceptance exception states the failed item, risk, temporary control, owner, due date, and whether public use is allowed. Safety failures cannot be accepted by commercial waiver.

Baseline

The final record captures firmware, model, configuration, site conditions, sample set, network status, and signatures. Later performance comparisons should reference this baseline.

SECTION 22

Preventive maintenance

Maintenance preserves safety, image quality, routing reliability, hygiene, and fill accuracy. Customer operators and GreenLoop technicians have different authorized tasks.

Interval	Task	Owner
Daily	Inspect lens, chamber, door, and visible damage	Site operator
Every 14 days	Clean lens and chamber; inspect liners and sensors	Site operator
Monthly	Functional test, jam review, and ventilation cleaning	Trained operator or service
Quarterly	Load-cell calibration and full safety test	GreenLoop certified technician
Annually	Electrical inspection, wear parts, anchors, and battery if fitted	GreenLoop certified technician

Cleaning

Use approved non-abrasive cleaner on a cloth. Do not spray into cameras, vents, displays, seams, or electrical areas. The unit must be placed in maintenance mode before chamber or bin cleaning.

Records

Completed work records include date, unit, task, findings, parts, software changes, test results, technician, and follow-up. Missing preventive maintenance can exclude a failure from the service-level calculation.

SECTION 23

Diagnostics and fault handling

Faults are classified by safety, service impact, and recovery path. The unit shows a user-safe message while detailed codes remain in the service interface.

Fault family	Typical response
Door or interlock	Block motion; inspect closure and sensor
Diverter position	Stop cycle; service inspection required
Camera quality	Clean lens; verify light and calibration
Weight instability	Remove item; inspect chamber and load cell
Bin full	Disable or redirect route according to policy
Network offline	Continue local sorting and queue events
Storage threshold	Prioritize upload and restrict image sampling

Remote diagnostics

Authorized support can view health, recent fault codes, software versions, and sanitized logs. Remote support access is time bound and audited. Safety resets that require physical inspection cannot be completed remotely.

Evidence preservation

Do not repeatedly clear a fault before logs are captured. For recurring classification issues, record site lighting, lens condition, item type, model version, confidence, route, and whether multiple items were present.

SECTION 24

Service levels and support

X1 Pro includes a monthly 99.5% availability target for the customer-facing cloud dashboard. Device local sorting availability and customer network availability are measured separately.

Severity	Example	Initial response target
P1 Critical	Safety risk or fleet-wide cloud outage	30 minutes, 24x7
P2 High	Site unable to sort with no workaround	2 business hours
P3 Medium	Degraded analytics or repeated non-safety fault	8 business hours
P4 Low	Question, report, or cosmetic issue	2 business days

Dashboard SLA

Availability equals eligible minutes minus unavailable eligible minutes, divided by eligible minutes. Planned maintenance announced at least 72 hours ahead, customer network failure, customer configuration, force majeure, and misuse are excluded as defined in the order.

Service credits

Eligible customers request review within 30 days. Credits are the sole financial remedy for an SLA miss unless the contract states otherwise.

Contractual dashboard target: 99.5% monthly availability for X1 Pro.

SECTION 25

Known limitations and exclusions

Known limitations guide deployment and interpretation. They are not permission to ignore a defect; unexpected behavior must still be recorded and assessed.

Condition	Potential effect	Recommended control
Multiple or overlapping items	Mixed appearance and wrong route	One-item prompts and retry
Food-soiled or wet items	Material features obscured	Route low confidence to Other
Lighting outside 300 to 1200 lux	Exposure and color shift	Correct site lighting
Dirty or fogged lens	Lower image quality	Inspect daily; clean every 14 days
Flexible film and strings	Entanglement or ambiguous shape	Exclude through signage
New regional packaging	Distribution shift	Collect approved samples and monitor
Closed bags	Contents hidden	Reject and ask user to separate

Field accuracy

No minimum field accuracy applies without an agreed site sample, operating range, maintenance compliance, software baseline, and measurement method. The Q3 business report may describe observed fleet performance but does not amend this specification.

Financial impact

This specification does not calculate revenue loss, customer churn, support cost, or waste-value impact caused by a limitation. Those claims require separately approved financial evidence.

SECTION 26

Pricing, licensing, and warranty

Prices below are FY2025 list prices before taxes, freight, installation, volume discounts, distributor terms, and contract-specific options. Recognized accounting revenue per unit will not necessarily equal list price.

Item	FY2025 list price
X1 Standard hardware	USD 6,800
X1 Pro hardware	USD 7,600
Standard cloud subscription	USD 149 per unit per month
Pro cloud subscription	USD 249 per unit per month
Standard installation	From USD 850 per site
Extended service	Quoted by region and fleet size

License

The customer receives a non-transferable right to use device software and subscribed cloud features during the applicable term. Models, firmware, service tools, and platform software are not sold as source code.

Warranty

The standard hardware warranty is 24 months from commissioning or 27 months from shipment, whichever occurs first. Consumables, misuse, unsupported waste, unauthorized modification, flooding, and missed maintenance are excluded.

SECTION 27

Compliance and sustainability

The X1 is designed using applicable electrical, machinery, electromagnetic compatibility, material, privacy, and accessibility requirements identified for the destination market. Certification scope is configuration and country specific.

Environmental design

- Replaceable camera, controller, motor, and sensor modules.
- Powder-coated steel structure with documented recycling path.
- Low-power idle state and scheduled display sleep.
- Remote diagnostics intended to reduce unnecessary service travel.
- Material analytics intended to support customer diversion programs.

Claims

Customers remain responsible for waste characterization, local bin labels, collection contracts, recycling acceptance, and sustainability claims. GreenLoop analytics estimate sorted activity; they do not prove final downstream recycling without external chain-of-custody evidence.

End of life

Units should be returned to GreenLoop or an approved partner for battery removal, data erasure, component recovery, and recycling. Customers must remove local labels and follow applicable e-waste law.

SECTION 28

Glossary and revision notes

This section defines terms used across the specification and records the principal changes in version 2.1.

Term	Definition
Field accuracy	Observed accuracy under documented customer-site conditions
Laboratory accuracy	Accuracy on the controlled validation set under specified conditions
Macro accuracy	Unweighted mean of accuracy across the five operational classes
SLA	Contractual service-level agreement and defined measurement method
Eligible minute	Minute included in the dashboard SLA calculation
Other	Operational fallback class; not a hazardous-waste determination
Multi-item event	Cycle in which more than one physical object is deposited

Version 2.1 changes

- Clarified single-item product scope and multi-item limitation.
- Added image retention configuration and administrative audit details.
- Updated laboratory accuracy to the approved GL-Vision 3.4.1 result.
- Clarified 99.5% dashboard SLA scope and exclusions.
- Added FY2025 list pricing and 24-month warranty baseline.

End of controlled document: PRD-GLX1-2025-v2.1. Printed copies are uncontrolled.